Class 17 Practice Assignment

Matlab Work-Up

Before summarizing the assignment, the file, `practice_17.m`, was created for performing the calculations. Within it a function of the same name was declared.

```
function practice_17()
% Calculations for the Class 17 Practice Assignment from REB, The Course
```

Assignment Summary

Reaction

$$A + B \rightarrow Y + Z \tag{1}$$

Rate Expression

$$r_1 = k_1 C_A \sqrt{C_B} \tag{2}$$

Reactor: adiabatic, gas-phase BSTR

Given Constants: ΔH_1 = –12.9 kcal mol^{-1}, $k_{0,1}$ = 1.75 x 10^9 L$^{0.5}$ mol$^{-0.5}$ s^{-1}, E_1 = 35.7 kcal mol^{-1}, $\hat{C}_{p,A}$ = 37 cal mol^{-1} K^{-1}, $\hat{C}_{p,B}$ = 34 cal mol^{-1} K^{-1}, $\hat{C}_{p,Y}$ = 36 cal mol^{-1} K^{-1}, $\hat{C}_{p,Z}$ = 38 cal mol^{-1} K^{-1}, V = 5.0 L, $y_{A,0}$ = 0.5, $y_{B,0}$ = 0.5, T_0 = 400 °C, P_0 1 atm, and $f_{A,f}$ = 0.5.

```
    %global constants available to all functions
    %given
    k01 = 1.75E9; %L^0.5 mol^-1.5 s^-1
    E1 = 35700; %cal/mol
    dH1 = -12900; %cal/mol
    CpA = 37.0; %cal/mol/K
    CpB = 34.0; %cal/mol/K
    CpY = 36.0; %cal/mol/K
```

```
CpZ = 38.0; %cal/mol/K
P0 = 1.0; %atm
T0 = 400 + 273.15; %K
yA0 = 0.5;
yB0 = 0.5;
fA = 0.5;
V = 5.0; %L
%known
Rpv = 0.08206; %L-atm/mol/K
R = 1.987; %cal/mol/K
%calculated
n0 = P0*V/Rpv/T0;
nA0 = yA0*n0;
nB0 = yB0*n0;
nAf = nA0*(1-fA);
```

Task: Analysis

- Quantities of Interest: t_f, and T_f

Preliminary Analysis

Only one reaction is taking place, and it is exothermic. The reactor is adiabatic, so the temperature is expected to increase as the reaction proceeds.

(Reactor Type) Model Function

BSTR Design Equations

$$\frac{dn_A}{dt} = -r_1 V \tag{3}$$

$$\frac{dn_B}{dt} = -r_1 V \tag{4}$$

$$\frac{dn_Y}{dt} = r_1 V \tag{5}$$

$$\frac{dn_Z}{dt} = r_1 V \tag{6}$$

$$\left(n_A \hat{C}_{p,A} + n_B \hat{C}_{p,B} + n_Y \hat{C}_{p,Y} + n_Z \hat{C}_{p,Z}\right) \frac{dT}{dt} - V\frac{dP}{dt} = -r_1 V \Delta H_1 \tag{7}$$

$$RT\left(\frac{dn_A}{dt} + \frac{dn_B}{dt} + \frac{dn_Y}{dt} + \frac{dn_Z}{dt}\right) + \left(n_A + n_B + n_Y + n_z\right) R\frac{dT}{dt} - V\frac{dP}{dt} = 0 \tag{8}$$

BSTR Model Variables: $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_Y$, $\underline{n}_Z$, $\underline{T}$, and $\underline{P}$

Additional Computable Unknowns: r_1, k_1, C_A, and C_B

$$k_1 = k_{0,1} \exp\left(\frac{-E_1}{RT}\right) \tag{9}$$

$$C_i = \frac{n_i}{V} \qquad i = A \text{ and } B \tag{10}$$

IVODE Solver Requirements

1. Initial values

$$t_{iv} = 0 \tag{11}$$

$$n_{A,iv} = n_{A,0} = y_{A,0}\frac{P_0 V}{RT_0} \tag{12}$$

$$n_{B,iv} = n_{B,0} = y_{B,0}\frac{P_0 V}{RT_0} \tag{13}$$

$$n_{Y,iv} = n_{Z,iv} = 0 \tag{14}$$

$$T_{iv} = T_0 \tag{15}$$

$$P_{iv} = P_0 \tag{16}$$

2. Stopping criterion

$$n_A = n_{A,f} = n_{A,0}\left(1 - f_{A,f}\right) \tag{17}$$

3. BSTR derivatives function (see below)

```
% BSTR reactor function
function [t, nA, nB, nY, nZ, T, P] = bstr_model_variables()
    %define the initial values
    ind_0 = 0;
    dep_0 = [nA0, nB0, 0, 0, T0, P0];

    %define the stopping criterion
    stopVar = 1;
    stopVal = nAf;

    %solve the BSTR design equations
    odes_are_stiff = true;
    [t, dep, success, message] = solve_ivodes(ind_0, dep_0, stopVar...
        , stopVal, @bstr_derivatives, odes_are_stiff);

    %check for solver issues
    if ~success
        disp('')
        disp(["BSTR model function issue: ", message])
        disp('Press any key to continue or CTRL-C to exit')
        pause;
    end

    %return the individual bstr model variables
    nA = dep(:,1);
    nB = dep(:,2);
    nY = dep(:,3);
    nZ = dep(:,4);
    T = dep(:,5);
    P = dep(:,6);
end
```

BSTR Derivatives Function

1. receives the BSTR model variables (t, n_A, n_B, n_Y, n_Z, T, and P) at the start of an integration step
2. calculates additional unknowns using equations (2), (9), and (10)
3. evaluates and returns design equation derivatives using equations (18), (19), and (20)

$$\underline{\underline{M}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sum_i \left(n_i \hat{C}_{p,i}\right) & -V \\ RT & RT & RT & RT & R\sum_i n_i & -V \end{bmatrix} \tag{18}$$

$$\underline{g} = \begin{bmatrix} -Vr_1 \\ -Vr_1 \\ Vr_1 \\ Vr_1 \\ -Vr_1\Delta H_1 \\ 0 \end{bmatrix} \tag{19}$$

$$\begin{bmatrix} \frac{dn_A}{dt} \\ \frac{dn_B}{dt} \\ \frac{n_Y}{dt} \\ \frac{n_Z}{dt} \\ \frac{dT}{dt} \\ \frac{dP}{dt} \end{bmatrix} = \underline{\underline{M}}^{-1}\underline{g} \tag{20}$$

```
%BSTR derivatives function
function ddt = bstr_derivatives(~, dep)
    %extract the dependent variables
    nA = dep(1);
    nB = dep(2);
    nY = dep(3);
    nZ = dep(4);
    T = dep(5);

    %calculate the additional unknowns
```

```
        k1 = k01*exp(-E1/(R*T));
        CA = nA/V;
        CB = nB/V;
        r1 = k1*CA*sqrt(CB);

        %form the mass matrix
        mm = zeros(6,6);

        %add the coefficients for the mole balances
        mm(1,1) = 1;
        mm(2,2) = 1;
        mm(3,3) = 1;
        mm(4,4) = 1;

        %add the coefficients for the energy balance
        mm(5,5) = nA*CpA + nB*CpB + nY*CpY + nZ*CpZ;
        mm(5,6) = -V;

        %add the coefficients for the differential gas law
        mm(6,1) = Rpv*T;
        mm(6,2) = Rpv*T;
        mm(6,3) = Rpv*T;
        mm(6,4) = Rpv*T;
        mm(6,5) = Rpv*(nA + nB + nY + nZ);
        mm(6,6) = -V;

        %form the right side vector
        rhs = [-r1*V; -r1*V; r1*V; r1*V; -r1*V*dH1; 0];

        %evaluate and return the derivatives
        ddt = mm\rhs;
    end
```

Items of Interest Function

1. Call the BSTR model function to get $\underline{t}$, $\underline{n}_A$, $\underline{n}_B$, $\underline{n}_Y$, $\underline{n}_Z$, $\underline{T}$, and $\underline{P}$
2. Extract the final time and temperature

$$t_f = \underline{t}\big|_{n_A = n_{A,f}} \tag{21}$$

$$T_f = \underline{T}\Big|_{n_A = n_{A,f}} \tag{22}$$

3. Tabulate, show, and save the results.

```
    % items of interest function
    function items_of_interest()
        %solve the BSTR design equations
        [t, ~, ~, ~, ~, T, P] = bstr_model_variables();

        %tabulate, show, and save the results
        item = ["Time"; "Temperature"];
        value = [t(end)/60; T(end) - 273.15];
        units = ["min"; "°C"];
        resultsTable = table(item, value, units);
        disp('')
        disp(resultsTable)
        disp('')
        writetable(resultsTable, 'results.csv');
    end

    % perform the calculations
    items_of_interest()
end
```

Results

The calculations were performed by calling the function, `practice_17`.

```
>> practice_17
        item          value     units
    -------------     ------    -----

    "Time"            7.2027    "min"
    "Temperature"     490.09    "°C"
```

Discussion

The calculations were performed as described. Reaching 50% conversion requires 7.2 min and results in a temperature of 490 °C. As noted above, the temperature is expected to increase

because the reaction is exothermic and the reactor is adiabatic.

The calculations were checked by noting that the reaction does not affect the total number of moles and the volume is constant, so the pressure depends only on the temperature. Solving the design equations yields the final pressure, and it can be calculated using the ideal gas law and the final temperature. Both calculations yielded the same value for the final pressure, 1.13 atm.

Summary

Reaching 50% conversion requires 7.2 min and results in a temperature of 490 °C.